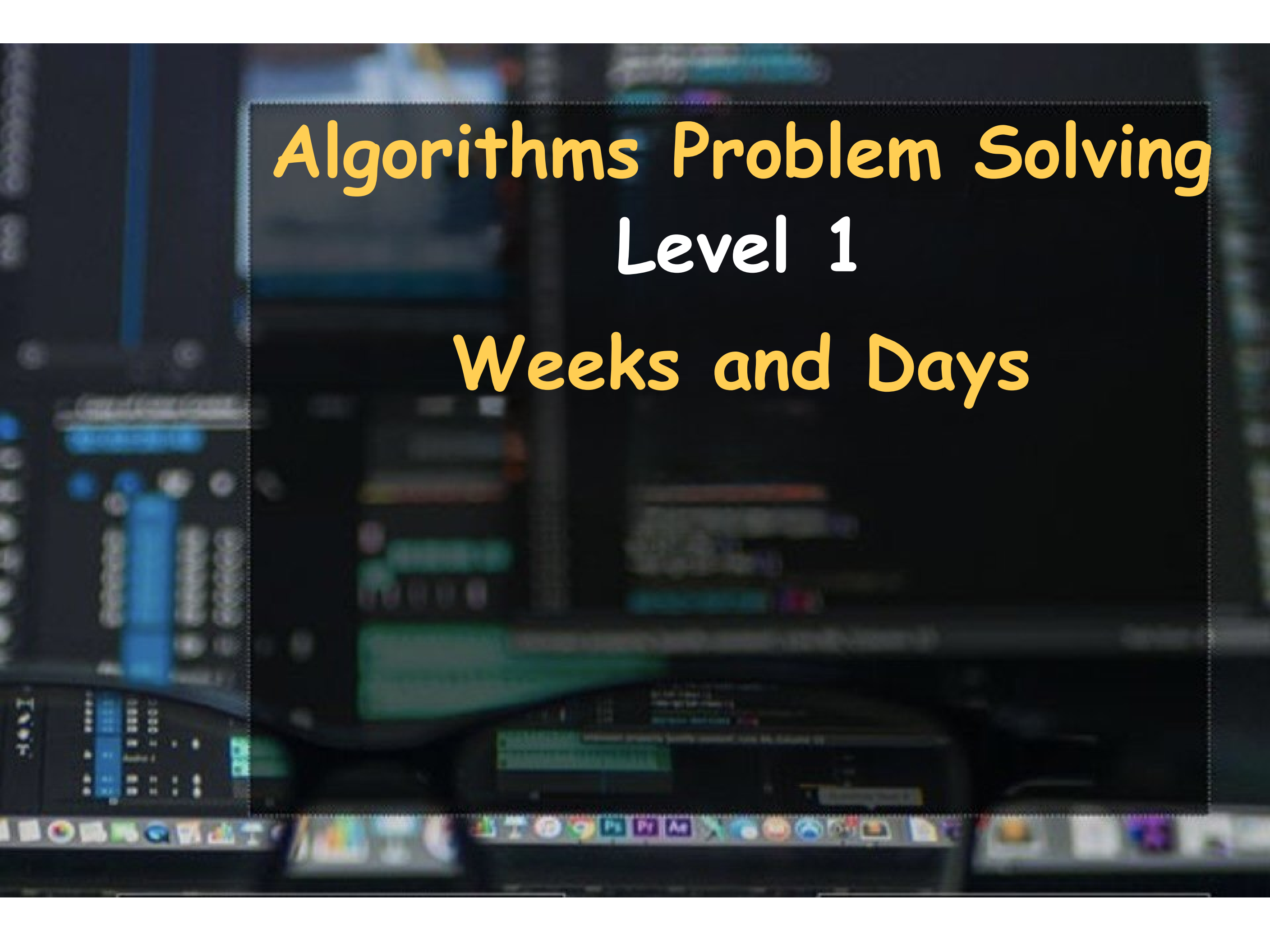




Algorithms Problem Solving

Level 1

Weeks and Days



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Definition

- ◆ This program reads a positive number of hours and converts it into days and weeks.
- ◆ The program uses conversion functions to calculate:
 - Total Days = Hours ÷ 24
 - Total Weeks = Days ÷ 7
- ◆ It also contains a function that can directly convert hours to weeks.

Explanation

- ◆ ReadPositiveNumb() keeps asking for the number of hours until the user enters a value greater than 0.
- ◆ HoursToDays() converts hours into days by dividing the hours by 24.
- ◆ HoursToWeeks() converts hours directly into weeks by dividing by ``24 × 7``.
- ◆ DaysToWeeks() converts days into weeks by dividing by 7.
- ◆ Printresult() displays the original hours, total days, and total weeks.
- ◆ In the current program, the weeks are calculated through:
``DaysToWeeks(HoursToDays(NumberOfHours))``
- ◆ This means the program first converts hours to days, then converts the days to weeks.

Steps in main()

1. Read a positive number of hours.
2. Convert the hours to days by dividing by 24.
3. Convert the calculated days to weeks by dividing by 7.
4. Print the hours, days, and weeks.
5. Return the result through the conversion functions.

💡 Example

◆ If the user enters:

365 hours

◆ Convert hours to days:

$365 \div 24 = 15.2083$ days

◆ Convert days to weeks:

$15.2083 \div 7 = 2.1726$ weeks

➡ Result

Total Hours = 365

Total Days = 15.2083

Total Weeks = 2.1726

Note

- ◆ One day = 24 hours.
- ◆ One week = 7 days = 168 hours.
- ◆ `HoursToWeeks()` can calculate weeks directly:
$$\text{NumberOfHours} / (24 * 7)$$
- ◆ In this program, `HoursToWeeks()` is defined but not used in `Printresult()`.

⊘ Warning

- ◆ The example description says `2.17 Weeks 15.20 Days`, but these are two different representations of the same 365 hours, not "2 weeks and 15.20 days."
- ◆ If the goal is to display the remaining days after complete weeks, additional calculation is needed.



Comparison

▨ Hours → Days → Weeks

- ✓ First divide hours by 24.
- ✓ Then divide days by 7.
- ✓ Used by ``Printresult()'`.

▤ Hours → Weeks directly

- ✓ Divide hours by ``24 * 7'`.
- ✓ Produces the same total number of weeks.
- ✓ Implemented by ``HoursToWeeks()'`.

✦ Summary

- ✓ 24 hours make one day.
- ✓ 7 days make one week.
- ✓ Hours can be converted to days and then to weeks.
- ✓ Functions can be combined by passing the result of one function to another.
- ✓ `HoursToWeeks()` provides a direct way to calculate weeks but is not used in the current output.

Thank you 😊